Goal

In this lab session, we get familiar with synchronous and asynchronous operations. And look at the difference in execution time when comparing the two.

(A)Synchronous operations

In the previous lab sessions we used synchronous operations this means that when the CPU starts an operation on the GPU it waits for said operation to finish before continuing the execution of the code. The CPU is thus unused for a period of time. When Asynchronous operations are used the CPU is able to perform another task while the GPU is busy. Note that the CPU is only able to perform another task that is not dependent on the results of the GPU task. To test the performance difference, the reduction technique from previous lab was used to calculate the minimum, maximum, sum and product of an array. Figure 1 shows how the code was structured to test the difference. In between each operation, a cudaDeviceSynchronize() was implemented in the synchronous test to ensure the CPU waited until the GPU was finished processing.

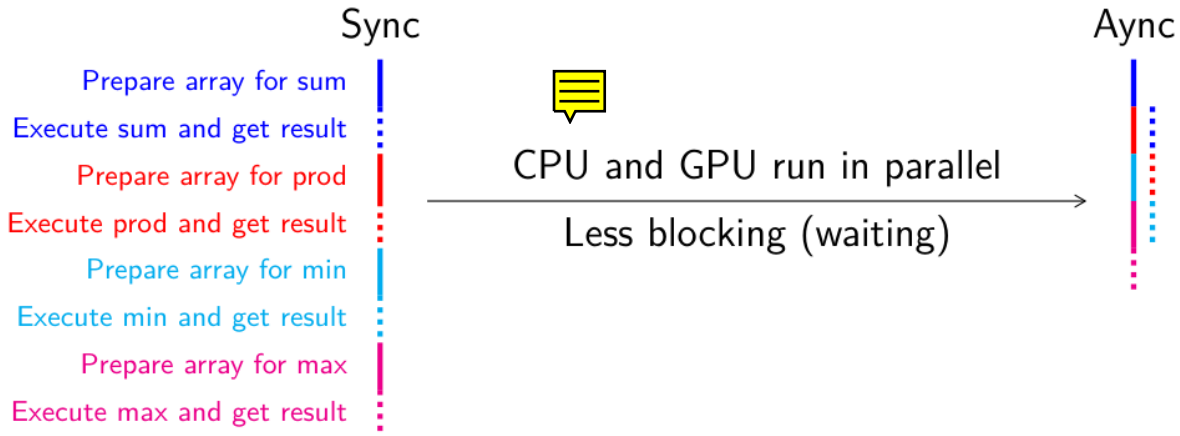


Figure 1: Synchronous and Asynchronous test program

The results in figure 2 show the execution time for the synchronous and asynchronous program for varying array sizes. From an array size of 256 to 1600 we can see a clear improvement when using asynchronous. It is unclear as to why the time improvement for asynchronous scheduling is lost at higher array sizes. Additionally a more complex array creation on the CPU was implemented to get a bigger impact of CPU execution time but no significant time differences were noted.

Conclusion

While theoretically Asynchronous operations are faster according to this test the time improvement is not always noticeable. The unexpected results could be due to the unbalanced execution times of the CPU vs the GPU. A more complex CPU calculation was added to improve the imbalance albeit without effect.

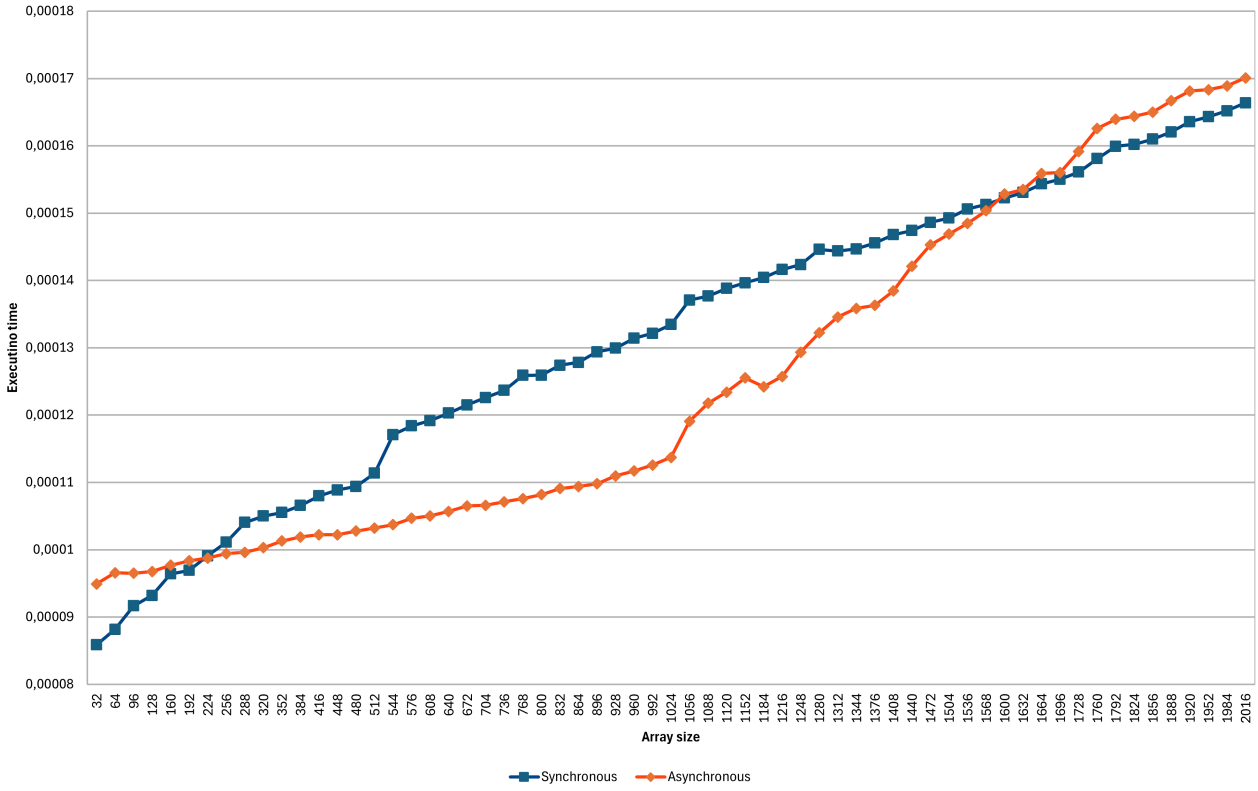


Figure 2: Test results